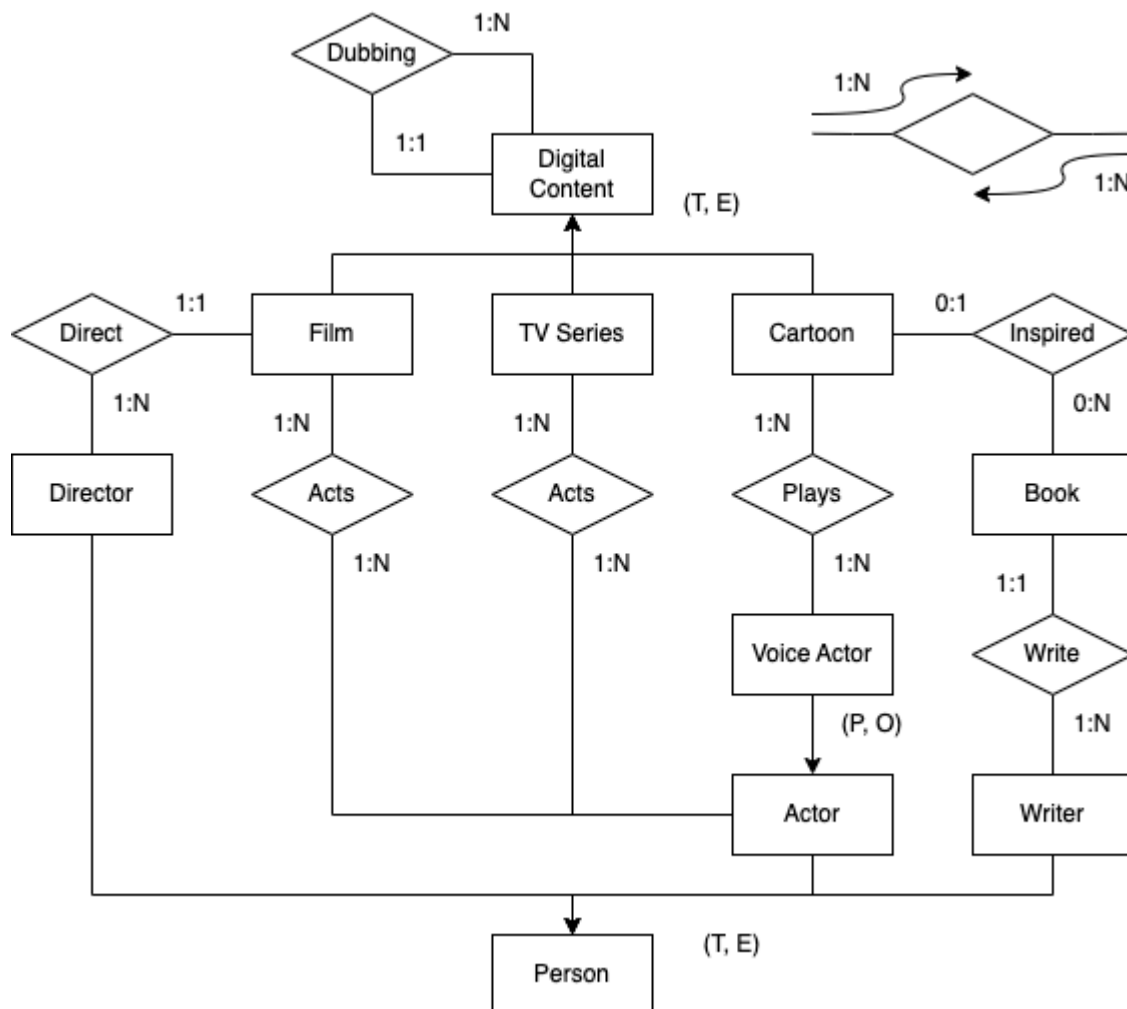


Last Name (Surname, Cognome) _____ First Name _____ Codice Persona _____

Consider the following ER Diagram, which represents a data structure for tracking Netflix's digital content.



The following attributes describe the entities. The primary keys are underlined.

- **Digital Content** - ID, title, duration, description, language, original (Y/N)
 - **Film** - (...), budget, number_oscars
 - **TV Series** - (...), number_episodes, season
 - **Cartoon** - (...), number_episodes, season
- **Book** - ID, name, number_volumes, publication_date
- **Person** - Personal ID, name, surname, birth_date, nationality, biography
 - **Actor** - (...), number_oscars
 - **Voice Actor** - (...), voice_type (high, low, etc.)
 - **Writer** - (...),
 - **Director** - (...), number_oscars

The following attributes describe the relationships.

- **Plays and Acts** - character_name

N.B. Only the original digital content can be dubbed. Actors are only associated with the original digital content.

Exercise 1 - Unstructured data models (1 PT)

Which DB technology would be the best to store the entities/rel. of the ER diagram? [you may not need all the rows of the table] (1 PT)

ENTITIES / REL.	DB TYPE	MOTIVATION

Exercise 2 - Neo4j (5 PT)

Consider the entities Person, Actor, Digital Content, Film, and TV Series from the ER model. Suppose you want to store the respective data instances in a graph database.

Sketch a graph model/example describing the nodes, main attributes, and edges. Either show an example graph or a graph with types. (1 PT)

Write a Cypher query to collect the total number of actors who acted in at least 10 Films that were dubbed in at least three other languages, including Italian. (2 PT)

Write a Cypher query to collect the original British Films and TV Series whose cast includes at least 10 non-British actors who also acted in a Japanese movie and won at least 1 Oscar (2 PT)

Exercise 3 - MongoDB (7 PT)

Consider the entities Voice Actor, Writer, Book, and Cartoon. How many collections would you define within a documental database (clearly state it)? How would you implement the relations between the concepts? Provide a simple documental representation. (1 PT)

Write a query to return the total number of voice actors who performed in each cartoon that was inspired by a book written by a writer named "Mark" (2 PT).

Write a query to return the total number of cartoons inspired by each book published after 17/03/2020 and whose cartoon includes at least one japanese voice actor (2 PT)

N.B. In this query only, be sure to check that at least one book inspired the movie (-0.5 PT otherwise).

Write a query to collect the total number of cartoons acted by each voice actor that was inspired by a book that includes at least 10 volumes and whose writer's name is "Bill" and was born after 10/10/2000 (2 PT)

Exercise 4 - Redis (4 PT)

Consider the Book entity and write the commands for the following operations.

Create a key-value pair with "best_book", the value "La ragazza delle Arance", and TTL equal to 100 (1 PT)

Check the TTL of the key-value pair "best_book" and make it permanently stored (1 PT)

Assume there exists a **set** named “HP Books” that includes all the Harry Potter books besides the first one. Add the movie “Harry Potter and the Philosopher Stone” to the set. (1 PT)

Check if the last movie “Harry Potter and the Deadly Hallows” is part of the set (1 PT)

Exercise 5 - SparkQL (3 PT)

Consider the following Director and Film tables.

<u>Personal ID</u>	Name	...

<i>Director ID</i>	<u>Film ID</u>	...

Write a SparkQL query to collect the total number of original films directed by directors whose nationality is equal to the film’s language (1.5 PT)

N.B. Consider the check between nationality and language works.

Write a SparkQL query to collect the total budget of films whose titles include the word “Curse” and whose director was born after 08/09/1994, for each director (1.5 PT)